

• Application of Derivative

> L-Hospital's Rule

(Use of derivative to find Limit)

Statement: If $f(x)$ and $g(x)$ be any two functions and $f'(x)$ and $g'(x)$ are continuous at a point $x=a$ and $f(a) = g(a) = 0$ then,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \frac{f'(a)}{g'(a)}, \quad g'(a) \neq 0.$$

Example: $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} \left(\frac{0}{0} \text{ form} \right)$

$$= \lim_{x \rightarrow 2} \frac{3x^2}{2x}$$

$$= \frac{3 \times 2^2}{2 \times 2}$$

$$= 3$$

Notes

① L-Hospital rule is applicable only for $\frac{0}{0}$ and

$\frac{\infty}{\infty}$ form.

Exercise 13.3

0 By using L-Hospital's rule, evaluate:

a) $\lim_{n \rightarrow 3} \frac{n^3 - 27}{n^2 - 9} \left(\frac{0}{0} \text{ form} \right)$

Using L-Hospital's rule,

$$\lim_{n \rightarrow 3} \frac{3n^2}{2n}$$

$$= \frac{3 \times 3^2}{2 \times 3}$$

$$= \frac{9}{2}$$

b) $\lim_{n \rightarrow a} \frac{n^n - a^n}{n - a} \left(\frac{0}{0} \text{ form} \right)$

using L-Hospital's rule.

$$\lim_{n \rightarrow a} \frac{n n^{n-1}}{1}$$

$$= n a^{n-1}$$

c) $\lim_{n \rightarrow 0} \frac{3n - \sin n}{n} \left(\frac{0}{0} \text{ form} \right)$

using L-Hospital's rule.

$$\lim_{n \rightarrow 0} \frac{3 - \cos n}{1}$$

$$= 3 - \cos 0$$

$$= 3 - 1$$

$$= 2$$

$$d) \lim_{n \rightarrow \infty} \frac{5n^2 + 4n - 3}{2n^2 - 3n + 5} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Using L-Hospital's rule

$$= \lim_{n \rightarrow \infty} \frac{10n + 4}{4n - 3}$$

$$= \lim_{n \rightarrow \infty} \frac{10 + \frac{4}{n}}{4 - \frac{3}{n}}$$

$$= \frac{10 + \frac{4}{\infty}}{4 - \frac{3}{\infty}}$$

$$= \frac{10}{4}$$

$$= \frac{5}{2}$$

$$= \frac{5}{2}$$

$$e) \lim_{n \rightarrow \infty} \frac{\ln(n^2 + 1)}{\ln(n^3 + 1)}$$

Using L-Hospital's rule,

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n^2 + 1} \times 2n}{\frac{1}{n^3 + 1} \times 3n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{2n(n^3 + 1)}{3n^2(n^2 + 1)}$$

$$= \lim_{n \rightarrow \infty} \frac{2n^3 + 2}{3n^3 + 3n}$$

$$= \lim_{n \rightarrow \infty} \frac{6n^2}{9n^2 + 3}$$

Using L-Hospital's rule

$$= \lim_{n \rightarrow \infty} \frac{12n}{18n}$$

Again using L-Hospital rule,

$$= \lim_{n \rightarrow \infty} \frac{12n}{18n} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Last, using L-Hospital rule,

$$= \lim_{n \rightarrow \infty} \frac{12}{18}$$

$$= \frac{2}{3}$$

∴

$$f, \lim_{n \rightarrow \infty} \frac{n^4}{e^n} \left(\frac{\infty}{\infty} \text{ form} \right)$$

∴ Using L-Hospital rule,

$$\lim_{n \rightarrow \infty} \frac{4n^3}{e^n} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Again, using L-Hospital rule,

$$= \lim_{n \rightarrow \infty} \frac{12n^2}{e^n} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Again, using L-Hospital rule,

$$= \lim_{n \rightarrow \infty} \frac{24n}{e^n} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Again, using L-Hospital rule,

$$= \lim_{n \rightarrow \infty} \frac{24}{e^n}$$

$$= \frac{24}{e^\infty}$$

$$= \frac{24}{\infty}$$

$$= 0$$

2.

$$a) \lim_{n \rightarrow 0} \frac{e^n + e^{-n} - 2\cos n}{\sin^2 n} \left(\frac{0}{0} \text{ form} \right)$$

using L-Hospital rule,

$$= \lim_{n \rightarrow 0} \frac{e^n - e^{-n} + 2\sin n}{2\cos n \times \sin n}$$

$$= \lim_{n \rightarrow 0} \frac{e^n - e^{-n} + 2\sin n}{\sin 2n} \left(\frac{0}{0} \text{ form} \right)$$

Again, Using L-Hospital rule,

$$= \lim_{n \rightarrow 0} \frac{e^n + e^{-n} + 2\cos n}{\cos 2n \times 2}$$

$$= \frac{e^0 + e^{-0} + 2\cos 0}{2\cos 2 \times 0}$$

$$= \frac{1 + 1 + 2}{2}$$

$$= \frac{4}{2}$$

$$= 2$$

$$b) \lim_{n \rightarrow 0} \frac{(e^n - 1) \tan n}{n^2} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{n \rightarrow 0} \left(\frac{e^n - 1}{n} \right) \cdot \left(\frac{\tan n}{n} \right)$$

$$= \lim_{n \rightarrow 0} \left(\frac{e^n - 1}{n} \right) \times 1$$

Using L-Hospital rule.

$$= \lim_{n \rightarrow 0} \frac{e^n}{1}$$

$$= e^0$$

$$= 1$$

$$c) \lim_{x \rightarrow 0} \frac{x - \tan x}{x^3} \left(\frac{0}{0} \text{ form} \right)$$

→ Using L-Hospital rule

$$= \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{3x^2}$$

Again, Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{-2 \sec x \times \sec x \cdot \tan x}{6x}$$

→ Again, Using L-Hospital rule.

$$= \lim_{x \rightarrow 0} -\frac{2}{3} \left[\sec^2 x \times \sec^2 x + \tan x \times \sec^2 x \cdot \tan^2 x \right]$$

$$= \lim_{x \rightarrow 0} -\frac{1}{3} (1 + 0)$$

$$= -\frac{1}{3}$$

$$d) \lim_{x \rightarrow 0} \frac{x - \sin x \cdot \cos x}{x^3} \left(\frac{0}{0} \text{ form} \right)$$

→ Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{1 - [\sin x \times (-\sin x) + \cos x \cdot \cos x]}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - [-\sin^2 x + \cos^2 x]}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2} \left(\frac{0}{0} \text{ form} \right)$$

Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{-(-\sin 2x) \times 2}{6x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 2x}{3x} \left(\frac{0}{0} \text{ form} \right)$$

Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{2 \cos 2x}{3}$$

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$$= \frac{2 \cos 0}{3}$$

$$= \frac{2}{3}$$

$$e) \lim_{n \rightarrow \frac{\pi}{2}} \frac{\sec 3n}{\sec 5n} \left[\frac{\infty}{\infty} \text{ form} \right]$$

~~Using L-Hospital rule~~

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{3 \sec 3n \cdot \tan 3n}{5 \sec 5n \cdot \tan 5n}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{1}{\frac{\cos 3n}{\cos 5n}}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{\cos 5n}{\cos 3n} \left[\frac{0}{0} \text{ form} \right]$$

Using L-Hospital rule,

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{-5 \cdot \sin 5n}{-3 \cdot \sin 3n}$$

$$= \lim_{n \rightarrow \frac{\pi}{2}} \frac{5 \sin 5n}{3 \sin 3n}$$

$$= \frac{5}{3} \frac{\sin 5\pi/2}{\sin 3\pi/2}$$

$$= \frac{5}{3} \times \frac{1}{-1}$$

$$= -\frac{5}{3}$$

$$f) \lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x} \left[\frac{0}{0} \text{ form} \right]$$

↳ Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{3 \tan^2 x} \times \frac{1}{\sec^2 x}$$

Again, Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{-(-\sin x)}{6 \tan x} \times \frac{1}{\sec^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{6 \tan x} \times \cos^2 x$$

• Again, Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{\cos x \times \cos^2 x + \sin x \times 2 \cos x \times (-\sin x)}{6 \sec^2 x}$$

$$= \frac{\cos 0 + \sin 0}{6 \sec^2 0}$$

$$= \frac{1}{6}$$

$$g) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan 5x}{\tan x} (\infty \text{ form})$$

↳ Using L-Hospital rule,

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{5 \cdot \sec^2 5x}{\sec^2 x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{5 \cos^2 5x}{\cos^2 x}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin 5x}{\cos x}$$

$$\frac{\sin x}{\cos x}$$

$$= \lim_{x \rightarrow \pi/2} \frac{\sin x \cdot \cos x}{\sin x \cdot \cos x} \left[\frac{0}{0} \text{ form} \right]$$

Using L-Hospital rule

$$= \lim_{x \rightarrow \pi/2} \frac{\sin x (-\sin x) + \cos x \times 5 \cos x}{\cos x \times \cos x + \sin x \times 5 (-\sin x)}$$

$$= \lim_{x \rightarrow \pi/2} \frac{-\sin x \cdot \sin x + 5 \cos x \cdot \cos x}{\cos x \cdot \cos x - 5 \sin x \cdot \sin x}$$

$$= \frac{-1 + 0}{0 - 5}$$

$$= \frac{-1}{-5}$$

$$= \frac{1}{5} \text{ Ans.}$$

$$h) \lim_{x \rightarrow 0^+} \frac{\ln \tan x}{\ln x} \left[\frac{\infty}{\infty} \text{ form} \right]$$

Using L-Hospital rule,

$$= \lim_{x \rightarrow 0^+} \frac{1}{\tan x} \times \sec^2 x$$

$$= \lim_{x \rightarrow 0} \frac{x \sec^2 x}{\tan x} \left[\frac{0}{0} \text{ form} \right]$$

Using L-Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{x \sec^2 x \cdot \tan^2 x + \sec^2 x}{\sec^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{\sec^2 x (x \tan^2 x + 1)}{\sec^2 x}$$

$$= 0 + 1$$

$$= 1 \text{ Ans.}$$

$$i) \lim_{n \rightarrow 0} \left[\frac{1}{n^2} - \frac{1}{\sin^2 n} \right] \quad [\infty - \infty \text{ form}]$$

$$= \lim_{n \rightarrow 0} \frac{\sin^2 n - n^2}{n^2 \sin^2 n} \quad \left(\frac{0}{0} \text{ form} \right)$$

Using ~~L~~-Hospital rule.

$$= \lim_{n \rightarrow 0} \frac{2 \sin n \cdot \cos n - 2n}{n^2 \times 2 \sin n \cdot \cos n + \sin^2 n \times 2n}$$

$$= \lim_{n \rightarrow 0} \frac{\sin 2n - 2n}{n^2 \times \sin 2n + 2n \sin^2 n} \quad \left[\frac{0}{0} \text{ form} \right]$$

Again using ~~L~~-Hospital rule,

$$= \lim_{n \rightarrow 0} \frac{2 \cos 2n - 2}{n^2 \times 2 \cos 2n + \sin 2n \times 2n + 2n \sin^2 n + \sin^2 n \times 2}$$

$$= \lim_{n \rightarrow 0} \frac{\sin^2 n - n^2}{n^2 \times \left(\frac{\sin n}{n} \right)^2 \times n^2}$$

$$= \lim_{n \rightarrow 0} \frac{\sin^2 n - n^2}{n^4} \quad \left[\frac{0}{0} \text{ form} \right] \quad \left[\because \frac{\sin n}{n} = 1 \right]$$

Using L-Hospital rule

$$= \lim_{n \rightarrow 0} \frac{2 \sin n \cdot \cos n - 2n}{4n^3}$$

Again, Using L-Hospital rule.

$$= \lim_{n \rightarrow 0} \frac{2 \cos 2n - 2}{12n^2}$$

Again, Using L-Hospital rule.

$$= \lim_{n \rightarrow 0} \frac{2 \times 2(-\sin 2n)}{24n}$$

Again Using L-Hospital rule.

$$= \lim_{n \rightarrow 0} \frac{-8 \cos 2n}{24}$$

$$= \frac{-8 \cos 0}{24}$$

$$= \frac{-1}{3}$$

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$$1. \lim_{n \rightarrow 0} \frac{ne^n - \ln(1+n)}{n^2} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{n \rightarrow 0} \frac{ne^n + e^n - \frac{1}{1+n}}{2n} \left[\text{using L-Hospital rule} \right]$$

$$= \lim_{n \rightarrow 0} \frac{e^n (1+n)^2 - 1}{2n(1+n)} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{n \rightarrow 0} \frac{e^n \times 2(1+n) + (1+n)^2 e^n}{2+4n} \left(\frac{0}{0} \text{ form} \right)$$

$$= \frac{e^0 \times 2(1+0) + (1+0)^2 e^0}{2+4 \cdot 0}$$

$$= \frac{3}{2}$$